

What is Express.js?

- Express.js is a lightweight and fast web framework for Node.js.
- It simplifies web application and REST API development.
- Built on top of Node.js.

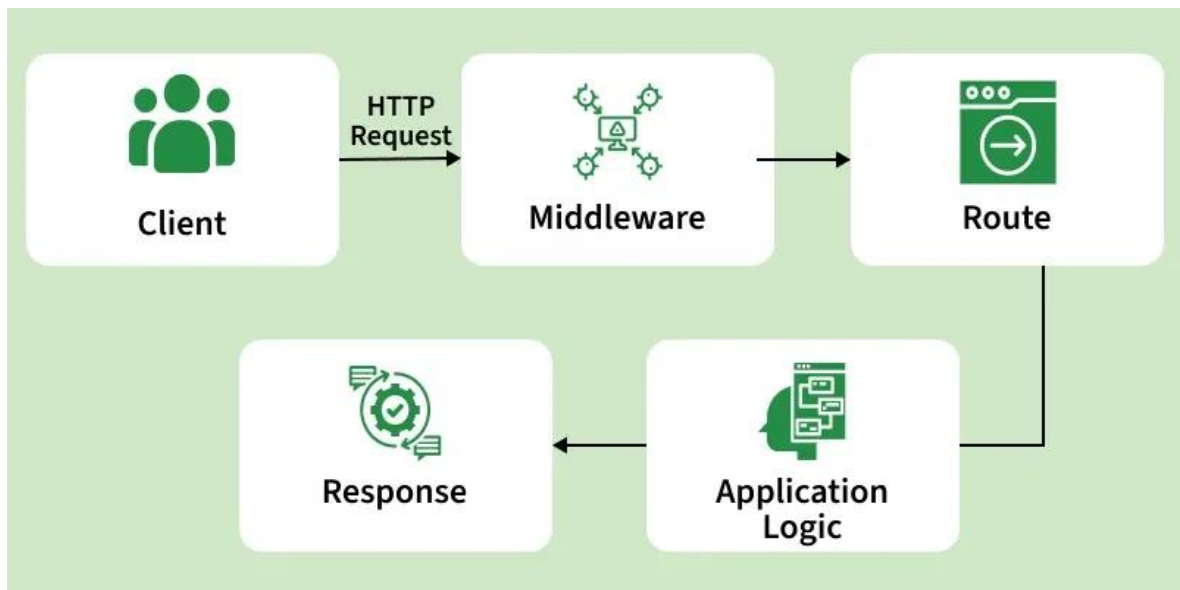
Why Do We Need Express.js?

Without Express.js, creating a server in Node.js requires writing more code using the built-in http module.

Express.js provides:

- Simple syntax
- Routing
- Middleware support
- Easy API development
- Better project structure

Architecture of Express.js



Installing Express.js

`npm install express`

What is Routing?

- Routing determines **how an application responds to a client's request**.
- Each route consists of:
 - URL (Path)
 - HTTP Method
 - Callback Function

Creating Routes

Method	Purpose
GET	Read Data
POST	Create Data
PUT	Update Entire Data
PATCH	Update Partial Data
DELETE	Delete Data

Route Parameters

Used to send values in the URL.

```
app.get("/students/:id", (req, res) => {  
  res.send(req.params.id);  
});
```

Query Parameters

Example `/students?name=Rahul&age=20`

```
app.get("/students", (req, res) => {  
  res.send(req.query);  
});
```

Middleware

Middleware is a **function that executes between the request and the response**. It can:

- Execute code
- Modify the request (req) or response (res)
- Perform authentication
- Log requests
- Handle errors
- Pass control to the next middleware using next()

Example

```
app.use((req, res, next) => {  
  console.log("Middleware Executed");  
  next();  
});
```

Built-in Middleware

Parse JSON

```
app.use(express.json());
```

Serve Static Files

```
app.use(express.static("public"));
```

Request Object (req)

Property	Description
req.params	Route Parameters
req.query	Query Parameters
req.body	Request Body
req.method	HTTP Method
req.url	Requested URL
req.headers	Request Headers

Response Object (res)

Method	Purpose
res.send()	Send Text or HTML
res.json()	Send JSON Data
res.status()	Set HTTP Status Code
res.redirect()	Redirect to Another URL
res.download()	Download a File

Send Text Response

```
app.get("/", (req, res) => {  
  res.send("Hello Students");  
});
```

Send JSON Response

```
app.get("/student", (req, res) => {
```

```
res.json({  
  id: 101,  
  name: "Rahul",  
  course: "BCA"  
});  
});
```

Send Status Code

```
app.get("/success", (req, res) => {  
  res.status(200).send("Success");  
});
```

Status Codes

Code	Meaning
200	OK
201	Created
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error

Folder Structure

```
express-app/  
|  
├─ node_modules/  
├─ routes/
```

- └─ controllers/
- └─ middleware/
- └─ models/
- └─ public/
- └─ package.json
- └─ server.js